

Analyze Operational Data to Improve Logistics & Warehousing Efficiency (LogiTech)



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Introduction

- ▶ **Logistics involves managing shipments, warehouses and inventory efficiently.**
- ▶ **Delays, high costs and poor stock planning can affect overall performance.**
- ▶ **The project focuses on understanding these logistics challenges through available business data.**
- ▶ **The analysis looks at delivery, warehouse, inventory and cost-related performance.**
- ▶ **The main aim is to identify problem areas and opportunities for improvement.**

Dataset Information

Dataset	Record	Primary Focus
 Shipment Delivery	209,402	Delivery Performance & Shipment Cost
 Warehouse Operation	19	Warehouse Efficiency & Utilization
 Inventory Stock Movement	2036	Inventory & Stock Management

❖ Key Data Attributes:

Shipment Data

- Delivery Mode
- Carrier
- Shipment Cost
- Delivery Status

Warehouse Data

- Utilization
- Operational Cost
- Inventory Turnover
- Order Fulfilment

Inventory Data

- Stock on Hand
- Reorder Level
- Carrying Cost
- Stockout Days

❖ Project Scope:

 3 Datasets

 211,457 Records

 19 Warehouses

 4 Analytics Tools

Project Objectives

Objective	Business Objective	Key Tasks	Primary Analytical Tool
O1: Improve Delivery Reliability & Cost Efficiency	Analyze shipment and delivery operations to identify delays, evaluate carrier performance, compare delivery methods and reduce shipment costs while ensuring timely deliveries.	• Understand shipment data • Assess data quality • Analyze delivery performance • Measure delivery KPIs • Compare carrier performance • Analyze delivery modes • Evaluate shipment costs • Build Excel Dashboard • Generate insights	Microsoft Excel
O2: Optimize Warehouse Capacity & Operational Efficiency	Analyze warehouse operations to improve utilization, processing efficiency and productivity.	• Understand warehouse data • Assess data quality • Analyze capacity • Measure utilization • Evaluate labour productivity • Analyze fulfilment • Study pick-pack time • Write SQL queries • Generate insights	MySQL

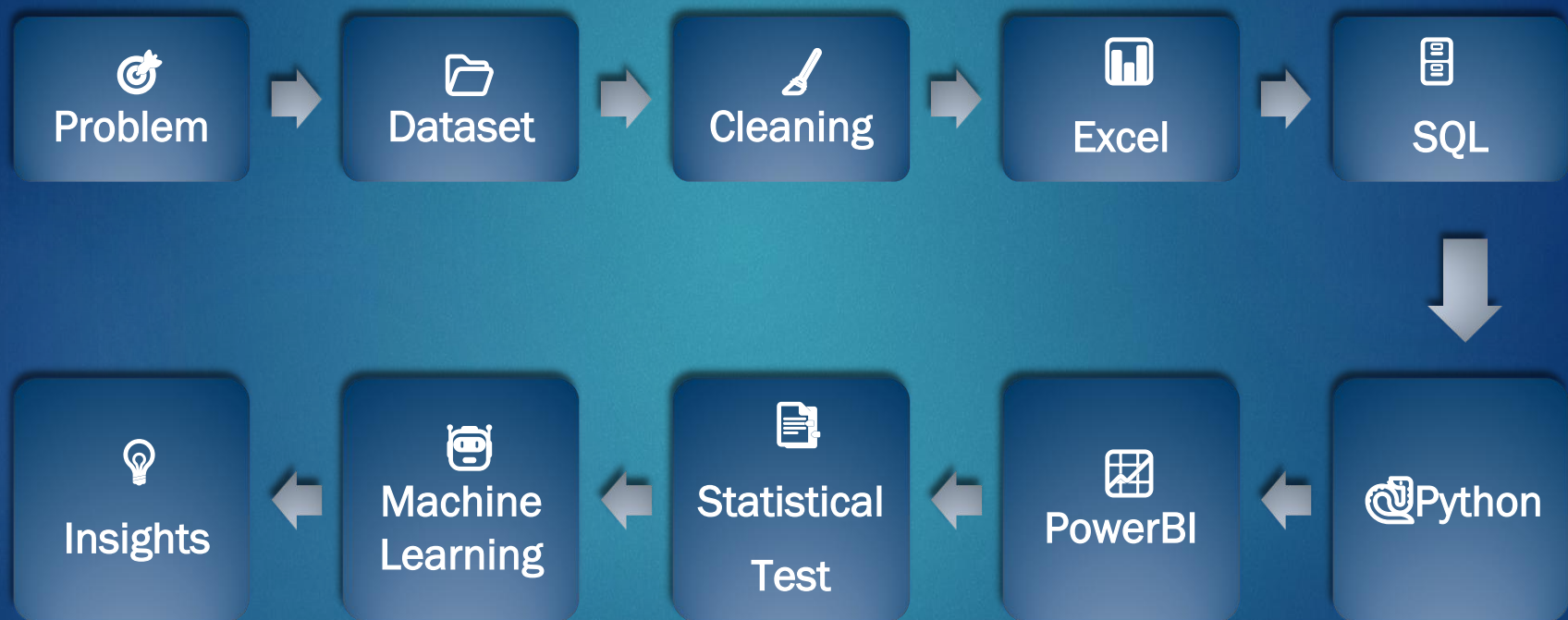
Project Objectives

Objective	Business Objective	Key Tasks	Primary Analytical Tool
O3: Strengthen Inventory Management	Analyze inventory to maintain optimal stock levels and improve reorder planning.	• Understand inventory data • Assess data quality • Perform EDA • Analyze stock availability • Study reorder trends • Analyze lead time • Analyze carrying cost • Create visualizations • Summarize insights	Python (Pandas, NumPy, Matplotlib)
O4: Maximize Financial Efficiency	Analyze logistics costs to identify cost drivers and recommend cost optimization strategies.	• Understand financial data • Assess data quality • Analyze shipment cost • Analyze operational cost • Compare warehouses • Design KPIs • Build Power BI Dashboard • Interpret findings	PowerBI

Project Objectives

Objective	Business Objective	Key Tasks	Primary Analytical Tool
O5: Perform Statistical Analysis	Analyze shipment and warehouse data to validate business assumptions and identify statistically significant relationships for better decision-making.	• Understand analytical problem • Prepare datasets • Perform Independent t-Test • Perform Pearson Correlation • Perform One-Way ANOVA • Perform Chi-Square Test • Interpret statistical results • Summarize business insights	Python (Pandas, SciPy, NumPy, Matplotlib)
O6: Predict Shipment Cost Using Machine Learning	Develop a predictive model to estimate shipment cost using shipment attributes and compare regression algorithms for accurate cost prediction.	• Prepare ML dataset • Feature selection & encoding • Train-Test Split (80:20) • Train Linear Regression • Train Random Forest Regressor • Compare model performance • Analyze feature importance • Interpret prediction results	Python (Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn)

Project Flow



Objective 1: Improve Delivery Reliability & Cost Efficiency

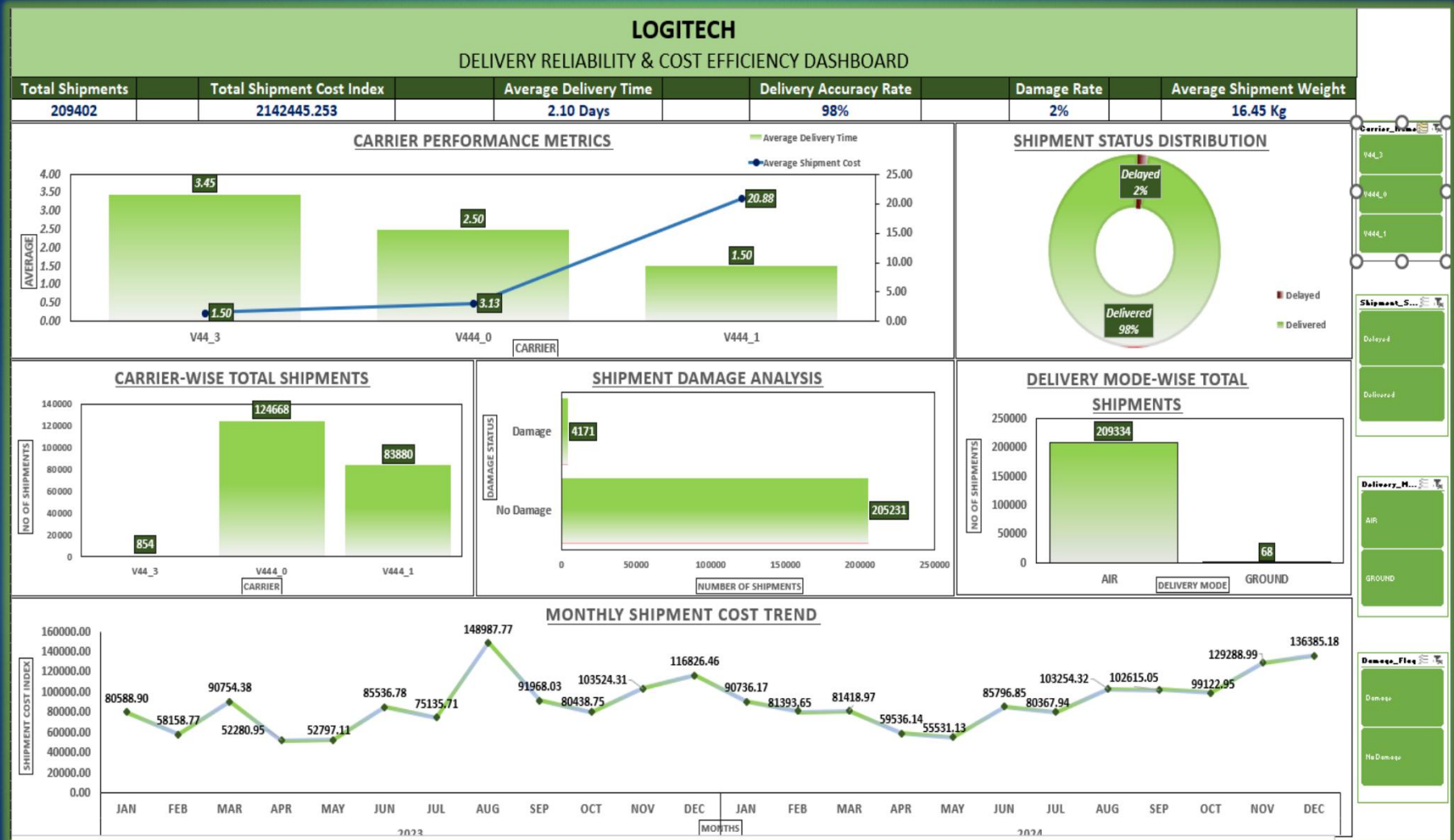


Fig: Delivery Reliability & Cost Efficiency Dashboard

Objective 1: Improve Delivery Reliability & Cost Efficiency

► Key Insights :

 209,402 shipments analyzed.

 Delivery accuracy achieved **98%**.

 AIR handled the majority of shipments.

 Shipment costs peaked during **August**.

 Damage rate remained around **2%**.

 V444_0 processed the highest shipment volume, while V444_1 had the highest average shipment cost.

► Business Outcome :

The Excel dashboard enables managers to monitor logistics performance, identify delivery trends, evaluate carrier efficiency, and support operational decision-making.

❖ Tools Used

- Microsoft Excel
- Pivot Tables
- Pivot Charts
- Slicers
- Conditional Formatting

❖ KPIs

- Total Shipments
- Shipment Cost Index
- Delivery Accuracy
- Damage Rate
- Average Delivery Time

Objective 2: Optimize Warehouse Capacity & Operational Efficiency

❖ Tasks :

- Analyze warehouse utilization
- Evaluate labour productivity
- Measure order fulfilment
- Study pick-pack efficiency

❖ Tools Used:

- MySQL
- SQL Queries
- Aggregate Functions
- CASE Statements
- Views
- ORDER BY & GROUP BY

Warehouse_ID	Warehouse_Capacity_Units	Current_Inventory_Units	Inventory_Turnover_Ratio	Order_Fulfillment_Rate	Avg_Pick_Pack_Time_Min	Warehouse_Utilization_Percentage	Labour_Hours_Per_Day	Operational_Cost_Per_Day
PLANT01	23990	18714	2.82	100.00	23.00	78.01	37.00	2122.08
PLANT02	3094	1780	2.82	100.00	25.00	57.53	3.00	169.99
PLANT03	22186	15904	61.92	98.14	28.00	71.68	31.00	1646.07
PLANT04	5820	3920	2.82	100.00	17.00	67.35	7.00	335.95
PLANT05	8631	6252	2.82	100.00	23.00	72.44	12.00	610.38
PLANT06	1098	859	2.82	100.00	20.00	78.23	3.00	95.19
PLANT07	5941	3228	2.82	100.00	25.00	54.33	6.00	239.79
PLANT08	855	462	21.87	82.19	29.00	54.04	3.00	48.31
PLANT09	650	401	2.99	100.00	18.00	61.69	3.00	37.30
PLANT10	2645	2128	2.82	100.00	29.00	80.45	4.00	210.07
PLANT11	7443	4578	2.82	100.00	29.00	61.51	9.00	508.38
PLANT12	10200	6921	5.97	100.00	16.00	67.85	13.00	1070.17
PLANT13	30142	16584	0.52	100.00	24.00	55.02	33.00	1557.92
PLANT14	12308	10154	2.82	100.00	29.00	82.50	20.00	1288.20
PLANT15	246	149	2.82	100.00	22.00	60.57	3.00	42.17
PLANT16	134299	69792	0.01	100.00	28.00	51.97	139.00	26797.44
PLANT17	179	112	2.82	100.00	24.00	62.57	3.00	9.61
PLANT18	2488	2041	2.82	100.00	24.00	82.03	4.00	831.20
PLANT19	156	104	2.82	100.00	100.00	66.67	3.00	13.31
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Fig: WarehouseOperations Dataset screenshot

❖ SQL Techniques Used:

- ✓ Data Validation
- ✓ Warehouse Capacity Analysis
- ✓ Warehouse Utilization Analysis
- ✓ Labour Productivity Analysis
- ✓ Order Fulfilment Analysis
- ✓ Pick-Pack Performance Classification

Objective 2: Optimize Warehouse Capacity & Operational Efficiency

Warehouse_ID	Warehouse_Capacity_Units	Current_Inventory_Units	Calculated_Utilization_Percentage
▶ PLANT14	12308	10154	82.50
PLANT18	2488	2041	82.03
PLANT10	2645	2128	80.45
PLANT06	1098	859	78.23
PLANT01	23990	18714	78.01
PLANT05	8631	6252	72.44
PLANT03	22186	15904	71.68
PLANT12	10200	6921	67.85
PLANT04	5820	3920	67.35
PLANT19	156	104	66.67
PLANT17	179	112	62.57
PLANT09	650	401	61.69
PLANT11	7443	4578	61.51
PLANT15	246	149	60.57
PLANT02	3094	1780	57.53
PLANT13	30142	16584	55.02
PLANT07	5941	3228	54.33
PLANT08	855	462	54.04
PLANT16	134299	69792	51.97

- Warehouse utilization ranges from **51.97% to 82.50%**.
- PLANT14** recorded the highest utilization (**82.50%**), indicating efficient use of storage capacity.

	Performance	Total_Warehouses
▶	Excellent	18
	Needs Improvement	1

- 18 warehouses** achieved **Excellent** fulfilment performance.
- PLANT08** was identified as **Needs Improvement**, with an order fulfilment rate of **82.19%**

	Warehouse_ID	Labour_Hours_Per_Day
▶	PLANT16	139.00
	PLANT01	37.00
	PLANT13	33.00
	PLANT03	31.00
	PLANT14	20.00
	PLANT12	13.00
	PLANT05	12.00
	PLANT11	9.00
	PLANT04	7.00
	PLANT07	6.00

- PLANT16** required the highest labour effort (**139 hours/day**).
- High labour requirements indicate potential opportunities for process optimization.

	Warehouse_ID	Avg_Pick_Pack_Time_Min	PickPack_Performance
▶	PLANT12	16.00	Average
	PLANT04	17.00	Average
	PLANT09	18.00	Average
	PLANT06	20.00	Average
	PLANT15	22.00	Average
	PLANT01	23.00	Average
	PLANT05	23.00	Average
	PLANT13	24.00	Average
	PLANT17	24.00	Average
	PLANT18	24.00	Average
	PLANT19	24.00	Average
	PLANT02	25.00	Average
	PLANT07	25.00	Average
	PLANT03	28.00	Needs Improvement
	PLANT16	28.00	Needs Improvement
	PLANT08	29.00	Needs Improvement
	PLANT10	29.00	Needs Improvement
	PLANT11	29.00	Needs Improvement
	PLANT14	29.00	Needs Improvement

- Several warehouses completed pick-pack operations within acceptable limits.
- Warehouses with pick-pack time above **25 minutes** were classified as **Needs Improvement**.

Objective 2: Optimize Warehouse Capacity & Operational Efficiency

	Average_Capacity	Minimum_Capacity	Maximum_Capacity
▶	14335.32	156	134299

Fig: Warehouse Capacity Summary

❖ Key Insights:

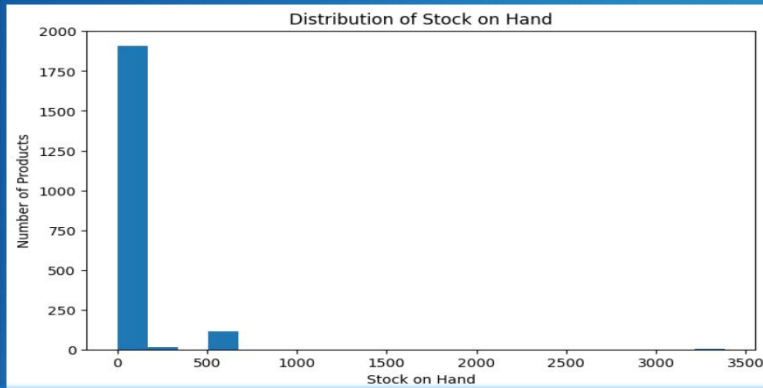
- Average warehouse capacity is **14,335 units**.
- Warehouse capacity ranges from **156 units** to **134,299 units**.
- Significant differences exist in warehouse size and operational workload.
- Warehouse utilization varies considerably across facilities.
- Pick-pack efficiency directly impacts warehouse productivity.

❖ Business Recommendations:

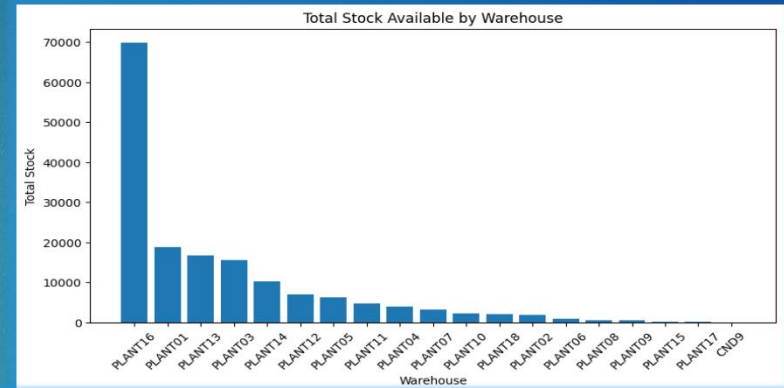
- Optimize inventory allocation across warehouses to improve capacity utilization.
- Improve pick-pack processes through layout optimization and automation.
- Continuously monitor warehouse utilization and fulfilment KPIs.
- Focus operational improvement initiatives on underperforming warehouses such as **PLANT08**.

Objective 3: Strengthen Inventory Management

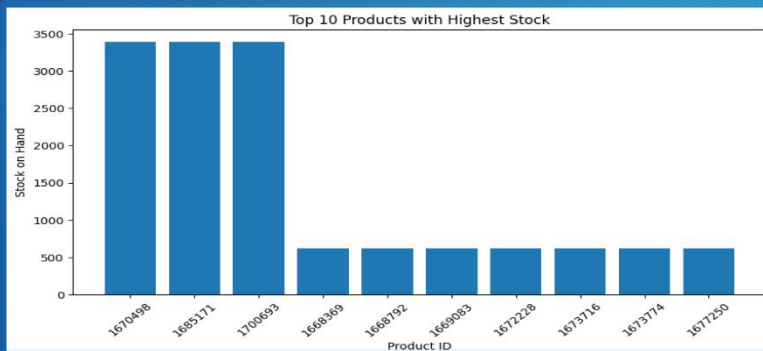
- **Task:** Analyze inventory availability across warehouses to identify stock distribution, Warehouse capacity utilization, and high-stock products for better inventory planning.



- Most products have relatively low stock quantities.
- Only a few products contain exceptionally high stock.



- PLANT16 has the largest inventory storage.
- Inventory is unevenly distributed across warehouses.



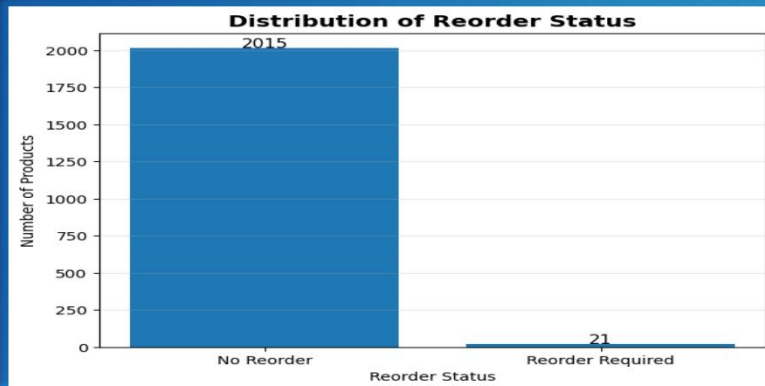
- A small number of products contribute to the highest inventory levels.
- These products require continuous monitoring to avoid excess stock.

❖ Key Findings:

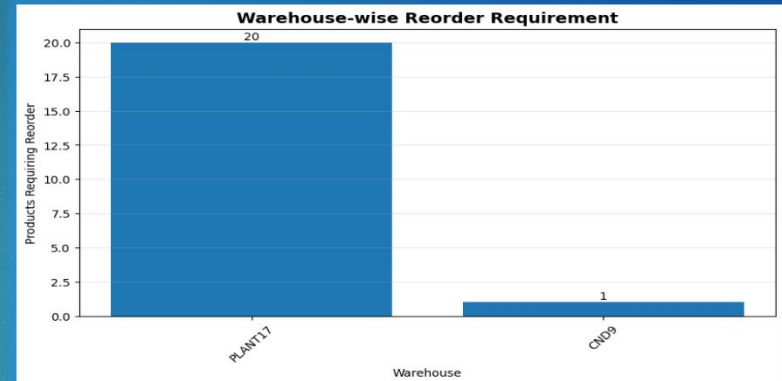
- Most products maintain low to moderate stock levels.
- **PLANT16** stores the highest inventory among all warehouses.
- A few products hold significantly higher stock than the remaining inventory.

Objective 3: Strengthen Inventory Management

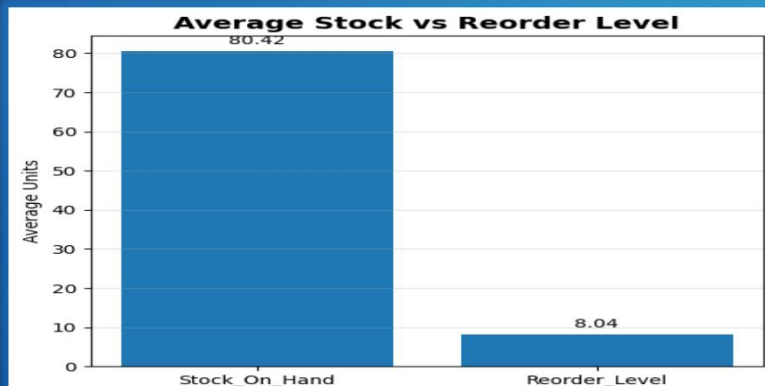
- Task: Identify products requiring replenishment and evaluate reorder requirements across warehouses.



- The majority of products do not require replenishment.
- Only a very small percentage needs immediate reorder.



- PLANT17 requires the maximum number of product replenishments.
- Most warehouses currently have sufficient inventory.



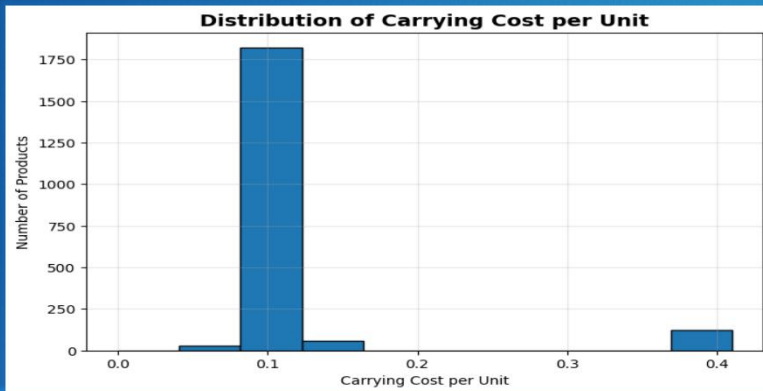
- Average stock on hand is much higher than the reorder threshold.
- Current inventory levels are generally adequate.

❖ Business Recommendations:

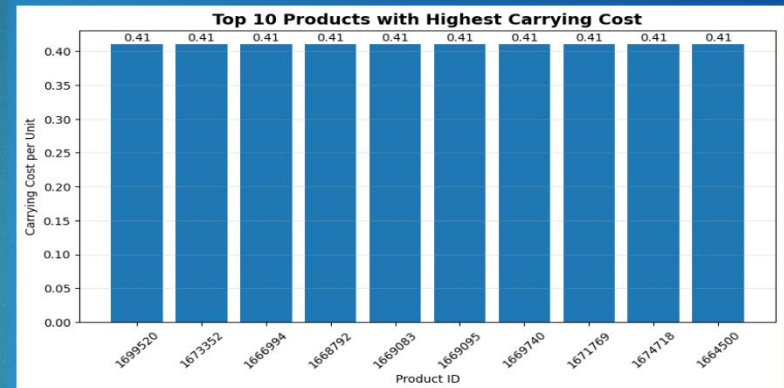
- Monitor PLANT17 inventory more frequently.
- Automate reorder alerts for products nearing reorder level.

Objective 3: Strengthen Inventory Management

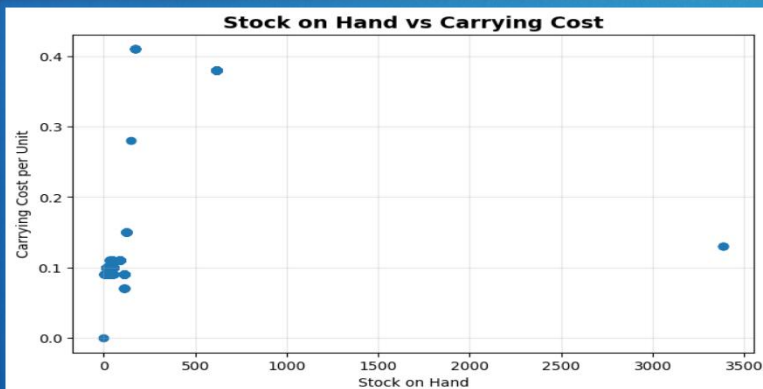
- Task: Analyze inventory carrying costs to identify high-cost warehouses and optimize inventory holding expenses.



- PLANT18 records the highest average carrying cost.
- Several warehouses maintain relatively low carrying costs.



- A few products generate the highest inventory holding expenses.
- These products should be reviewed for inventory optimization.



- No strong relationship exists between stock quantity and carrying cost.
- Carrying cost is influenced by multiple operational factors.

❖ Business Recommendations:

- Reduce excess inventory in high-cost warehouses.
- Improve inventory turnover to lower holding costs.
- Review stocking strategy for high carrying-cost products.

Objective 4: Maximize Financial Efficiency

- ❖ **Task :** Provide an executive-level overview of shipment, warehouse, and financial performance using integrated KPIs.

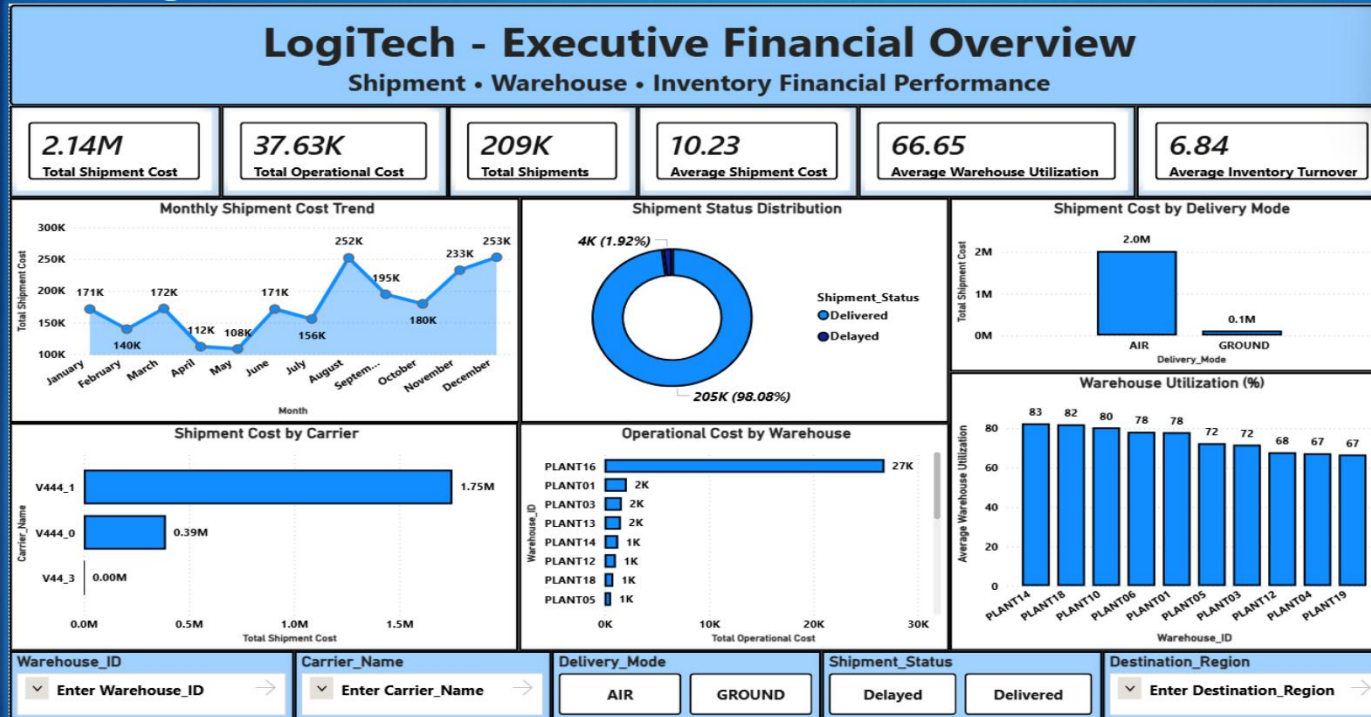


Fig: Executive Financial Overview Dashboard

❖ KPIs:

- Total Shipment Cost
- Total Operational Cost
- Total Shipments
- Average Shipment Cost
- Warehouse Utilization
- Inventory Turnover

❖ Key Insights :

- Total shipment cost exceeds 2.14M.
- Delivery accuracy remains around 98%.
- AIR transportation dominates shipment volume.
- PLANT16 incurs the highest warehouse operational cost.
- Carrier V444_1 contributes the largest shipment cost.

Objective 4: Maximize Financial Efficiency

- ❖ **Task :** Evaluate warehouse efficiency using operational cost, warehouse utilization, inventory turnover, and reorder analysis.

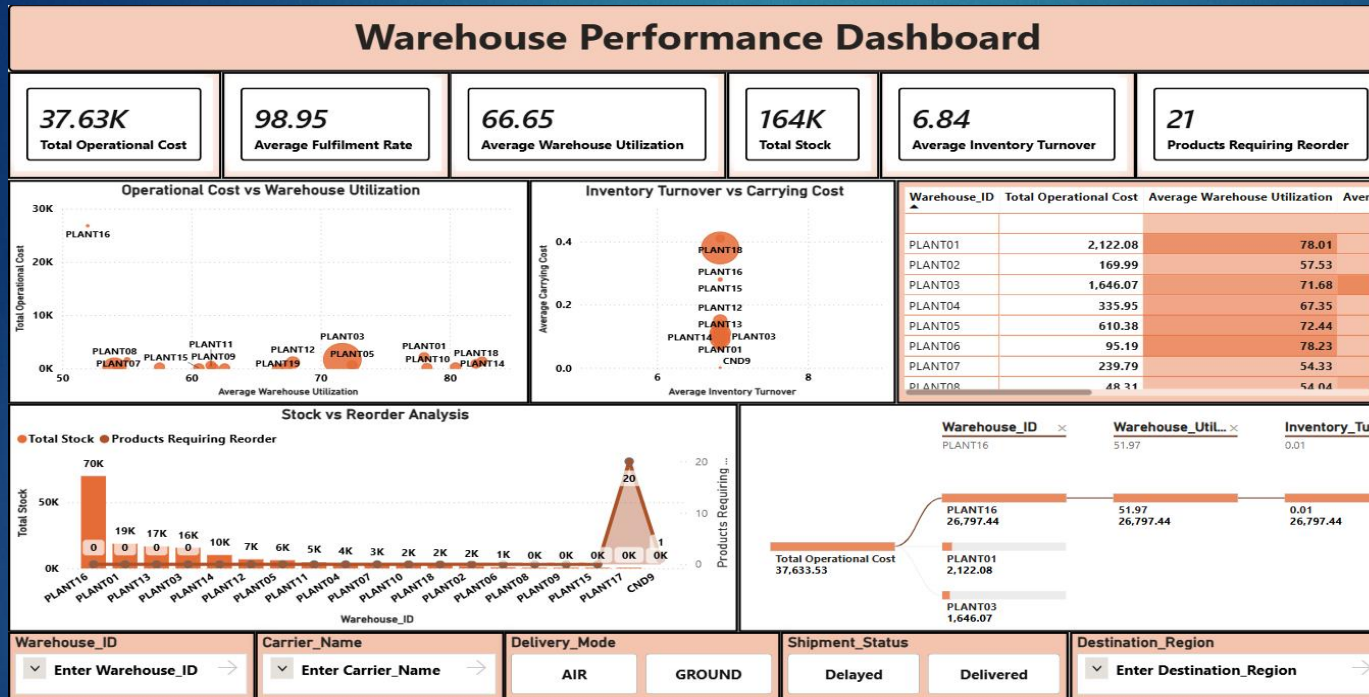


Fig: Warehouse Performance Dashboard

❖ KPIs:

- Total Operational Cost
- Average Fulfilment Rate
- Warehouse Utilization
- Total Stock
- Inventory Turnover
- Products Requiring Reorder

❖ Key Insights :

- PLANT16 has the highest operational cost.
- Warehouse utilization varies significantly across plants.
- Only **21 products** require immediate reorder.
- Most warehouses maintain excellent fulfilment performance.

Objective 4: Maximize Financial Efficiency

- ❖ **Task :** Analyze shipment costs, warehouse operational expenses, and inventory carrying costs to identify cost drivers and recommend strategies for improving logistics efficiency.



Fig: Cost Optimization Dashboard

❖ Key Insights :

- Air transportation contributes the highest shipment cost.
- Shipment cost peaks during August, November, and December.
- PLANT18 and PLANT16 have the highest carrying costs.
- Key Influencers identify factors driving higher shipment costs

❖ KPIs:

- Total Shipment Cost
- Cost per Shipment
- Operational Cost
- Average Carrying Cost
- Reorder Percentage
- Inventory Turnover

Objective 5: Validate Key Business Insights

Test	Business Objective	Result	Decision	Business Interpretation
Independent Sample t-Test	Compare shipment cost between Air and Ground deliveries.	$p = 0.000$	Reject H_0	Shipment cost differs significantly between Air and Ground transportation. Delivery mode influences logistics cost.
Pearson Correlation	Analyze relationship between Warehouse Utilization and Operational Cost.	$r = -0.313$ $p = 0.192$	Fail to Reject H_0	Weak negative relationship exists. Warehouse utilization alone does not significantly affect operational cost.

Objective 5: Validate Key Business Insights

Test	Business Objective	Result	Decision	Business Interpretation
One-Way ANOVA	Compare shipment cost across different carriers.	$F = 756.96$ $p = 0.000$	Reject H_0	Shipment cost varies significantly among carriers. Carrier selection has a direct impact on transportation cost.
Chi-Square Test	Check association between Delivery Mode and Shipment Status.	$\chi^2 = 0.507$ $p = 0.477$	Fail to Reject H_0	Delivery mode and shipment status are independent. Delays are influenced by operational factors rather than transport mode.

Objective 5: Validate Key Business Insights

❖ Key Business Outcomes :

- ✓ Delivery mode affects shipment cost.
- ✓ Carrier choice impacts logistics cost.
- ✓ Warehouse utilization does not directly determine operational cost.
- ✓ Shipment delays are not strongly linked to delivery mode.
- ✓ Key findings were checked before making recommendations.

Objective 6: Shipment Cost Prediction Using Machine Learning

❖ Tasks :

- ❑ Develop a Machine Learning regression model to predict shipment cost using shipment information.
- ❑ Compare different regression algorithms to identify the best-performing model for logistics cost estimation.

❖ Dataset Used:

- ❑ **ShipmentDelivery_Cleaned**
 - Total Records: 209,402
 - Target Variable: **Shipment_Cost**

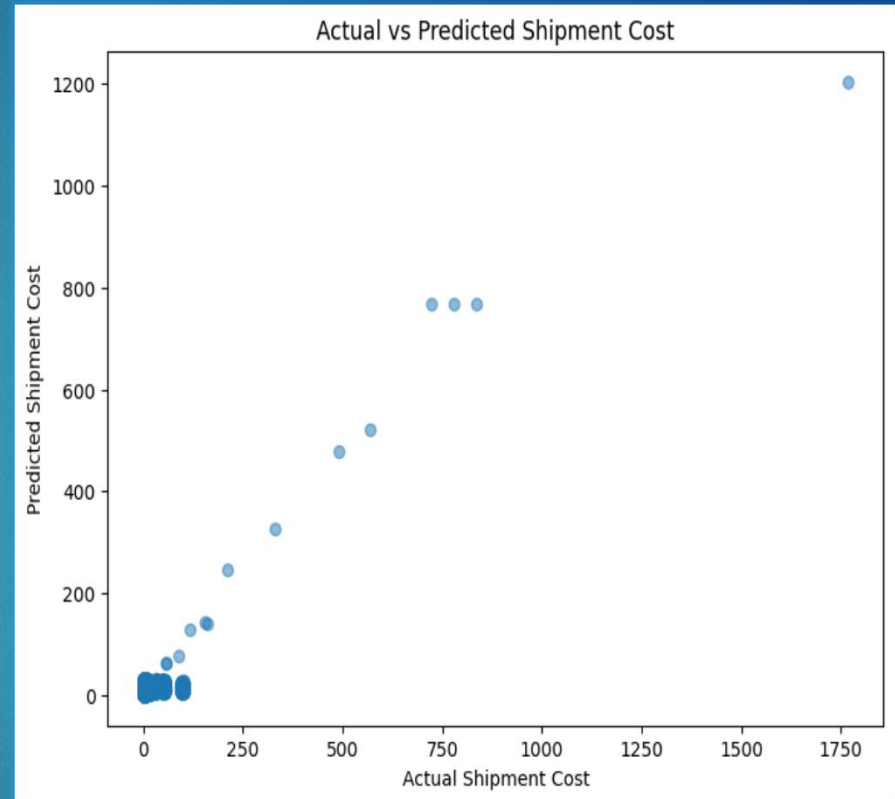
❖ Model Performance Summary

Model	MAE	RMSE	R ² Score
Linear Regression	15.36	46.13	-2.57
Random Forest	10.78	20.40	0.30

Objective 6: Shipment Cost Prediction Using Machine Learning

index	Feature	Importance
0	Weight	0.94044
1	Carrier_Name	0.01115
2	Delivery_Mode	0.03175
3	Warehouse_ID	6.53001808827711e-05
4	Destination_Region	0.0

	Model	MAE	RMSE	R2 Score
0	Linear Regression	15.46	46.1251	-2.569027
1	Random Forest	10.95	20.7748	0.275982



❖ Key Findings :

- Random Forest outperformed Linear Regression in predicting shipment cost.
- Shipment Weight was identified as the most influential factor affecting shipment cost.
- Delivery Mode and Carrier Name also contributed to the prediction.
- The model achieved lower prediction errors using Random Forest.

Project Summary

Module	Status
 Excel Dashboard	✓ Completed
 SQL Analysis	✓ Completed
 Python EDA	✓ Completed
 Power BI Dashboard	✓ Completed
 Statistical Analysis	✓ Completed
 Machine Learning	✓ Completed

❖ Project Coverage:

- ☐ Studied shipment, warehouse and inventory activities.
- ☐ Checked delivery performance and shipment costs.
- ☐ Examined warehouse capacity, utilization and productivity.
- ☐ Studied stock levels, reorder needs and inventory costs.
- ☐ Reviewed major operational and financial cost areas.
- ☐ Checked important business assumptions using data.
- ☐ Estimated shipment costs using past shipment information.

Conclusion

- ❑ Identified key areas affecting delivery, warehouse and inventory performance.
- ❑ Highlighted warehouses, shipments and inventory areas requiring attention.
- ❑ Identified major cost-related areas and opportunities for improvement.
- ❑ Provided data-based insights to support better operational planning.
- ❑ Generated actionable recommendations to improve logistics operations and decision-making.



THANK YOU

Any Questions.....??